

THE UNIVERSITY OF DODOMA



COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CIVE DAY 2026 — PROJECT PRESENTATION

ACADEMIC YEAR: 2025/2026

PROJECT TITLE:

**A DIGITAL PLATFORM FOR INTERNSHIP STUDENT TEACHER MANAGEMENT
SYSTEM**

MEMBERS

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1. INTRODUCTION

a. Background

In Tanzania, student teachers from universities and colleges are required to complete a practical internship in secondary and primary schools across the country. This involves placing students in schools, assigning supervisors (assessors), monitoring teaching activities through daily logbooks, and issuing completion certificates at the end of the internship period. Coordination spans multiple stakeholders: students, school assessors, head teachers, District Education Officers (DEOs), and university administration. Traditionally, this entire process has been handled manually — through paper logbooks, physical assessment forms, and in-person coordination — leading to significant inefficiencies.

b. Problem Statement

The manual management of student teacher internships creates the following critical challenges:

- No centralized system to track hundreds of student teachers across regions and districts.
- Paper logbooks are prone to loss, forgery, and inconsistency in record-keeping.
- Assessors have no efficient digital tool to record evaluations and observations.
- DEOs and head teachers lack real-time visibility into student progress.
- Certificate issuance is delayed due to multi-level manual approval processes.
- Communication between university admin, schools, and students is slow and unreliable.

c. Main Objective

To design and develop a web-based system that digitizes and automates the management of student teacher internship in Tanzania.

Specific Objectives:

- Automate student registration, school selection, and subject assignment.
- Enable digital logbook submission with assessor feedback and remarks.
- Provide assessors with tools to evaluate and grade student performance.
- Give DEOs and head teachers a dashboard for district-wide progress monitoring.
- Generate tamper-proof, digitally-signed completion certificates and official letters.
- Integrate AI tools to assist students in creating Schemes of Work and Lesson Plans.

2. THE CORE INNOVATION

a. System Architecture Overview

"The system follows a Monolithic Architecture based on Django's MVT (Model-View-Template) pattern. All modules — authentication, student management, assessor evaluation, DEO dashboard, and AI tools — are developed and deployed as a single unified application, containerized using Docker for isolated and reproducible deployment."

ACCESS LAYER: Web Browser (HTTPS)				
Student Teacher Assessor Admin DEO Head Teacher				
APPLICATION TIER: Django Web Application (Python 3.12 / Django 5.2 — MVT Architecture)				
AUTH MODULE Login / Register Password Reset	STUDENT MODULE School Selection Logbook / Reports	ASSESSOR MODULE Evaluations Remarks / Grades	BOARD / DEO Dashboard Approvals	ADMIN MODULE User Management Assignments
AI TOOLS Scheme of Work Lesson Plans	PDF ENGINE Certificates Logbook PDFs	SMS / EMAIL Africa's Talking Gmail SMTP	CACHING Redis Session Store	REST APIs AJAX Endpoints JSON Responses
DATA LAYER				
PostgreSQL 16 Database (Schools / Students / Assessments / Logbooks)		Docker Container (Web + DB Isolated)	External Services - OpenAI / Google Gemini (AI) - Gmail SMTP (Email)	

b. Methodology

Development Methodology: Agile Iterative Model

This project used the Agile Iterative Development Methodology. The system was built in short iterations, each producing a working part of the system. After each iteration, the team reviewed progress, gathered feedback from stakeholders, and improved the next iteration. This approach allowed the team to add new features gradually, fix issues early, and deliver a complete and tested internship management system at the end.

Each iteration followed this cycle: Plan → Design → Implement → Test → Review → Next Iteration.

Technical Tools and Technologies Used

Programming Language	Python 3.12
Web Framework	Django 5.2 (MVT Architecture)
Database	PostgreSQL 16 (via Docker)
AI Integration	OpenAI GPT-4 & Google Gemini — AI-powered Scheme of Work & Lesson Plan generation
PDF Generation	ReportLab — Certificates, Logbooks, Official Letters with watermarks & signatures
Containerization	Docker & Docker Compose — isolated, reproducible deployment
Frontend	HTML5, CSS3, JavaScript — Responsive multi-role web interface
Communication	Gmail SMTP (email) · Africa's Talking API (SMS notifications to head teachers)
Version Control	Git — source code management and collaboration
Testing	Django TestCase (unit tests) · Manual functional testing per iteration